

Field and Galois Theory

Labix

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Abstract

References

- Abstract Algebra by David S. Dummit and Richard M. Foote
- Field and Galois Theory by P. Morandi

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1 Field Extensions

1.1 Basics of Field Extensions

Definition 1.1.1 (Field Extensions)

Let F be a field. A field extension of F is a field K and a ring homomorphism $\phi : F \rightarrow K$. We denote it as K/F or $F < K$.

Recall that since homomorphisms between fields are always injective, we lose nothing by simply considering field extensions K/F as subfields.

Lemma 1.1.2

Let F be a field. Let K/F be a field extension. Then K is a vector space over F .

Proof Define the action of F on K by simply left multiplication. Then the axioms of a vector space is satisfied through properties of F as a field. ■

Definition 1.1.3 (Degree of an Extension)

Let F be a field. Let K/F be a field extension. Define the degree of the field extension to be

$$[K : F] = \dim_F(K)$$

Proposition 1.1.4 (The Tower Law) Let $F < K < E$ be field extensions. Then

$$[E : F] = [E : K][K : F]$$

Moreover, if $\{a_i | i \in I\}$ is a basis for E over K and $\{b_j | j \in J\}$ is a basis for K over F then

$$\{a_i b_j | i \in I, j \in J\}$$

is a basis for E over F .

Definition 1.1.5 (Finite Extensions) Let K/F be a field extension. We say that K/F is a finite extension if $[K : F]$ is finite.

1.2 Finitely Generated Field Extensions

Definition 1.2.1 (The Smallest Subfield Containing a Set)

Let F be a field. Let K/F be a field extension. Let $X \subseteq K$. Define the smallest subfield of K containing F and X to be

$$F(X) = \text{Frac}(F[X])$$

Proposition 1.2.2

Let F be a field. Let K/F be a field extension. Let $X \subseteq K$. Then

$$F(X) = \bigcap_{\substack{R \text{ is a field} \\ F \cup X \subseteq R \subseteq K}} R$$

Definition 1.2.3 (Finitely Generated Field Extensions)

Let F be a field. Let K/F be a field extension. We say that K/F is finitely generated if there exists $a_1, \dots, a_n \in K$ such that

$$K = F(a_1, \dots, a_n)$$

Proposition 1.2.4

Let F be a field. Let $f \in F[x]$ be a polynomial. Let K/F be a field extension of f such that f contains a root $a \in K$. Then there is a field isomorphism

$$F(a) \cong \frac{F[x]}{(f)}$$

Proposition 1.2.5 Every finite field extension is finitely generated.

Proof Let K/F be a finite field extension, say $n = [K : F]$. Then by definition, $K = \text{span}\{k_1, \dots, k_n\}$ where $k_i \in K$ for each i . Hence every element of K can be written as a finite linear combination of $k_1, \dots, k_n$ over F . Thus $K \subseteq F(x_1, \dots, x_n)$. Since K is a field that contains F and $\{k_1, \dots, k_n\}$, by proposition 9.4.6, we have that $F(k_1, \dots, k_n) \subseteq K$ and so we conclude. ■

The converse to the above is not true.

Example 1.2.6

The field extension $\mathbb{Q}(\pi)/\mathbb{Q}$ is a finitely generated field extension but not finite.

Definition 1.2.7 (Simple Extensions)

Let F be a field. Let K/F be a field extension. We say that K/F is a simple extension if there exists $a \in K$ such that

$$K = F(a)$$

1.3 Algebraic Elements

Recall the notion of algebraic elements and the minimal polynomials from groups and rings.

Definition 1.3.1 (Algebraic and Transcendental Elements) Let $F < K$ be fields. Then $a \in K$ is said to be algebraic over F if there exists a non zero polynomial $f \in F[x]$ such that $f(a) = 0$. Otherwise it is said to be transcendental.

Definition 1.3.2 (Minimal Polynomial) Let $F < K$ be field extensions and $a \in K$ be algebraic over F . Then the minimal polynomial of a over F is the monic irreducible polynomial $p(x) \in F[x]$ of least degree such that $p(a) = 0$. It is often denoted $\min(F, a)$.

A polynomial can be the minimal polynomial for one of its roots, but not for its other roots. For example, the minimal polynomial of $2^{1/3}$ is $x^3 - 2$, but it is not the minimal polynomial for $2^{1/3}e^{2\pi i/3}$ even though it is a root of $x^3 - 2$. On the other hand, $x^2 + 1$ is the minimal polynomial of both i and $-i$.

Proposition 1.3.3 Let $F < K$ be fields and $a \in K$ be algebraic over F . Then the following are true.

- $\min(F, a)$ exists and is unique.
- If a is a root of $g \in F[x]$, then $f|g$.

Proposition 1.3.4 Let $F < K$ be a field extension and $a \in K$ algebraic over F . Let $\deg(\min(F, a)) = n$. Then the following are true.

- There is a field isomorphism

$$F(a) \cong \frac{F[x]}{(\min(F, a))}$$

- The set $\{1, a, \dots, a^{n-1}\}$ is a basis for $F(a)$ over F .

Proof Suppose that f is the minimum polynomial. We prove that $L : F[x]_{<n} \rightarrow F[x]/(f)$ defined by $L(g) = g + (f)$ is a field isomorphism. It is easy to check that it is a F -linear map. For injectivity, suppose that $L(v) = 0$. Then $v \in (f)$ as well as $\deg(v) < \deg(f)$ by construction forces

v to be 0. For surjectivity, suppose that $r + (f) \in F[x]/(f)$ and $\deg(r) < \deg(f)$. Then we can just choose $r \in F[x]_{<n}$ and we are done. Thus L is an isomorphism.

Using the fact that $F[x]/(f) \cong F(a)$, we can construct a basis for $F(a)$ from L . But L has the basis $\{1, x, \dots, x^{n-1}\}$. Transporting it through the field isomorphisms, we get $\{1, a, \dots, a^{n-1}\}$ is a basis of $F(a)$. ■

Example 1.3.5

The element $\sqrt{2}$ is not contained in $\mathbb{Q}(\sqrt{3})$.

Proof If it is, then $\mathbb{Q}(\sqrt{2})$ is an intermediate field of $\mathbb{Q}(\sqrt{3})/\mathbb{Q}$. By the tower law, we have

$$3 = [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}(\sqrt{2})]$$

which is a contradiction. ■

Example 1.3.6

We have $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Proof

Clearly we have $\mathbb{Q}(\sqrt{2} + \sqrt{3}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$. It remains to show that $[\mathbb{Q}(\sqrt{2} + \sqrt{3}), \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3})] = 4$. Notice that $\sqrt{2} + \sqrt{3}$ satisfies the polynomial $x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$. By the generalized Gauss's lemma, this element is irreducible in $\mathbb{Q}[x]$ if and only if it is irreducible in $\mathbb{Z}[x]$. By Eisenstein's criterion with $P = (2)$, we conclude that $x^4 - 10x^2 + 1$ is irreducible. Then we have $[\mathbb{Q}(\sqrt{2} + \sqrt{3}), \mathbb{Q}] = \deg(\min(\mathbb{Q}, \sqrt{2} + \sqrt{3})) = 4$. ■

Proposition 1.3.7

Let K be a field extension of F . If each $a_1, \dots, a_n \in K$ is algebraic over F , then $F(a_1, \dots, a_n)$ is a finite dimensional vector space with

$$[F(a_1, \dots, a_n) : F] \leq \prod_{k=1}^n [F(a_k) : F]$$

1.4 Algebraic Extensions

Every field extension can be classified into an algebraic extension or a transcendental extension or a mixture of both. Since $\mathbb{R}$ contains both algebraic elements and transcendental elements of $\mathbb{Q}$, it is the prototypical example of such a field extension. Algebraic extensions will be studied briefly here, while more is said on transcendental extensions in field and Galois theory.

Definition 1.4.1 (Algebraic Extensions)

Let K/F be a field extension. We say that K/F is an algebraic extension if every element of K is algebraic over F .

In particular, notice that $a \in K$ is an algebraic element over F if and only if $F(a)$ is an algebraic extension.

Proposition 1.4.2

Let F be a field. Let K/F be a field extension. Then the following are true.

- If K/F is finite, then K/F is algebraic.
- If K/F is algebraic and L/K is algebraic, then L/F is algebraic.
- $a \in K$ is algebraic over F if and only if $[F(a) : F]$ is finite.

Since finitely generated field extensions is a finite concatenation of simple extensions, we can ask the question of whether the simplicity criteria can be relaxed to finitely generated extensions. The following proposition shows that it is possible.

Proposition 1.4.3

Let F be a field. Let K/F be a finitely generated field extension. Then K/F is finite if and only if K/F is algebraic.

If one knows of colimits, then it is easy to see that algebraic extensions are just colimits of finite extensions. This means that algebraic extensions are just finite extensions applied a countable number of times.

Definition 1.4.4 (Algebraically Closed Fields)

Let F be a field. We say that F is algebraically closed if F has no algebraic extensions of F .

Proposition 1.4.5

Let F be a field. Then the following are equivalent.

- F is algebraically closed.
- There are no finite extensions of F .
- Every polynomial in F factors completely into linear factors in F .
- Every polynomial in F has a root in F .
- Every irreducible polynomial over F has degree 1.

Proposition 1.4.6

Let F be a field. Then there exists a unique (up to isomorphism) algebraically closed field K such that K/F is a field extension.

Definition 1.4.7 (Algebraic Closure)

Let F be a field. Define the algebraic closure $\overline{F}$ of F to be the unique algebraically closed field containing F .

Proposition 1.4.8

Let F be a field. Let K/F be an algebraic field extension. Then the following are true.

- There exists an embedding $i : K \rightarrow \overline{F}$ such that $i(a) = a$ for all $a \in F$.
- The above embedding can be extended to an isomorphism $\overline{K} \cong \overline{F}$.

2 More on Field Homomorphisms

2.1 Homomorphisms and Fixed Fields

Definition 2.1.1 (F -Homomorphisms)

Let F be a field. Let K/F and L/F be field extensions. Let $\varphi : K \rightarrow L$ be a homomorphism. We say that φ is an F -homomorphism if $\varphi(a) = a$ for all $a \in F$.

Definition 2.1.2 (The Set of F -Homomorphisms)

Let F be a field. Let K/F and L/F be field extensions. Define the set of F -homomorphisms to be

$$\text{Hom}_F(K, L) = \{\varphi : K \rightarrow L \mid \varphi \text{ is an } F\text{-homomorphism}\}$$

Definition 2.1.3 (The Group of F -Automorphisms)

Let F be a field. Let K/F be a field extension. Define the group of F -automorphisms of K to be the set

$$\text{Aut}_F(K) = \text{Hom}_F(K, K)$$

together with composition of functions.

Proposition 2.1.4

Let F be a field. Let K/F be a field extension such that $K = F(X)$ is finitely generated for some $X \subseteq K$. Let $\varphi_1, \varphi_2 \in \text{Aut}_F(K)$ such that $\varphi_1|_X = \varphi_2|_X$. Then $\varphi_1 = \varphi_2$.

Lemma 2.1.5

Let F be a field. Let K/F be a finite field extension. Then $\text{Aut}_F(K)$ is a finite group.

2.2 Permuting the Roots of a Polynomial

Proposition 2.2.1

Let F be a field. Let K/F and L/F be field extensions. Let $\varphi \in \text{Hom}_F(K, L)$ be an F -automorphism. If $f \in F[x]$ and $a \in F$ is a root of f , then $\varphi(a)$ is a root of f .

Proposition 2.2.2

Let F be a field. Let K/F be a field extension. Let $f \in F[x]$ be an irreducible polynomial. Let $\alpha, \beta \in K$. Then there exists a unique F -homomorphism $\varphi : F(\alpha) \rightarrow F(\beta)$ such that $\varphi(\alpha) = \beta$.

Proposition 2.2.3

Let F be a field. Let K/F be a field extension. Let $f \in F[x]$ be an irreducible polynomial. Let $\alpha \in K$ be a root of f . Then we have

$$|\text{Hom}_F(F(\alpha), K)|$$

Proposition 2.2.4

Let F be a field. Let K/F be a finite field extension. Let L/F be a field extension. Then we have

$$|\text{Hom}_F(K, L)| \leq [K : F]$$

2.3 The Fixed Field of Automorphisms

Definition 2.3.1 (Fixed Fields) Let K be a field. Let $S \subseteq \text{Aut}(K)$ be a subset of automorphisms. Define the fixed field of S to be

$$\text{Fix}_K(S) = \{a \in K \mid \tau(a) = a \text{ for all } \tau \in S\}$$

Lemma 2.3.2 Let K be a field. Let $S \subseteq \text{Aut}(K)$. Then $\text{Fix}_K(S) < K$ is a subfield.

Proposition 2.3.3

Let K be a field and let $G \leq \text{Aut}(K)$ be a finite subgroup. Then

$$[K : \text{Fix}_K(G)] = |G|$$

Proof Let $G = \{\sigma_1, \dots, \sigma_n\}$. Suppose that $a_1, \dots, a_{n+1} \in K$. Set $L = K^G < K$, the fixed field of G . Let

$$v_i = \begin{pmatrix} \sigma_1(a_i) \\ \vdots \\ \sigma_n(a_i) \end{pmatrix} \in K^n$$

for $i = 1, \dots, n+1$. Since K^n is a vector field over K with dimension n , $v_1, \dots, v_{n+1}$ are linearly dependent with

$$c_1 v_1 + \dots + c_k v_k = 0$$

the shortest linear independence relation for $c_i \in K \setminus \{0\}$. Without loss of generality suppose that $x_1 = 1$ and $\sigma_1 = \text{id}_K$. Write the relation as

$$(v_1 \quad \dots \quad v_n) \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

This means that $(x_1, \dots, x_k)^T$ is a non-trivial solution of the n simultaneous linear equations. For any $\sigma \in G$, applying σ to the system of linear equation only permutes the rows of the matrix so it leaves exactly the same set of equations, but the solution is now $(\sigma(x_1), \dots, \sigma(x_k))^T$. But this must be the same solution as before. If not, then $\sigma(x_1) = \sigma(1) = 1 = x_1$ implies that we can get a shorter linear dependence relation which is a contradiction. Thus we have that $\sigma(x_i) = x_i$ for each $i = 1, \dots, k$ and for every $\sigma \in G$. So each $x_i \in K^G = L$ together with the first row of the system of linear equations give a non-trivial L -linear relation among the a_i .

Thus every K^G -linearly independent set of elements of K has size less than $|G|$, proving that $[K : K^G] \leq |G|$.

By theorem 1.1.6, we have that

$$|G| \leq |\text{Aut}_{K^G}(K)| = |\text{Emb}_{K^G}(K, K)| \leq [K : K^G]$$

and so we conclude. ■

Example 2.3.4

Let $\sigma : \mathbb{C}(x) \rightarrow \mathbb{C}(x)$ be the field homomorphism $x \mapsto x^{-1}$. Let $\tau : \mathbb{C}(x) \rightarrow \mathbb{C}(x)$ be the field homomorphism $x \mapsto e^{2\pi i/n} x$. Then

$$\text{Fix}_{\mathbb{C}(x)}(\langle \sigma, \tau \rangle) = \mathbb{C}(x^n + x^{-n})$$

Proof

Notice that $\langle \sigma, \tau \rangle \cong D_{2n}$. Let L be the fixed field. By the above, we have $[\mathbb{C}(x) : L] = 2n$. Let $t = x^n + x^{-n}$. Then notice that $x \in \mathbb{C}(x)$ satisfies the polynomial equation $y^{2n} - y^n t + 1 \in \mathbb{C}(t)[y]$. Hence $\mathbb{C}(x)$ is an algebraic extension of $\mathbb{C}(t)$ that is finitely generated, and so it is a finite extension. Hence $[\mathbb{C}(x) : \mathbb{C}(t)] \leq 2n$. On the other hand, $\mathbb{C}(t)$ is a subfield of $\mathbb{C}(x)$, and t is fixed by σ and τ . Hence $t \in L$ and $\mathbb{C}(t) < L$. By the tower law, we have

$$2n \geq [\mathbb{C}(x) : \mathbb{C}(t)] = [K : L][L : \mathbb{C}(t)] = 2n[L : \mathbb{C}(t)]$$

Hence $[L : \mathbb{C}(t)] = 1$ and so $\mathbb{C}(x^n + x^{-n})$ is the desired fixed field. ■

3 Normal and Separable Extensions

3.1 The Splitting Field of a Polynomial

Definition 3.1.1 (Splitting a Polynomial)

Let F be a field. Let K/F be a field extension. Let $f \in F[x]$. We say that f splits over K if f factors completely into linear factors in $K[x]$.

Proposition 3.1.2

Let F be a field. Let $f \in F[x]$ be a polynomial. Then the following are true.

- There exists a field extension K/F such that K contains a root of f and $[K : F] \leq \deg(f)$.
- There exists a field extension K/F such that f splits over K and $[K : F] \leq \deg(f)!$.

Definition 3.1.3 (Splitting Fields)

Let F be a field. Let K/F be a field extension. Let $f \in F[x]$. We say that K is a splitting field of f if the following are true.

- There exists $f \in F[x]$ such that f splits over K .
- If L/F is a field extension such that f splits over L , then $\deg(L) = \deg(K)$.

Example 3.1.4

The splitting field of $x^3 - 2 \in \mathbb{Q}[x]$ is the field $\mathbb{Q}(2^{1/3}, e^{2\pi i/3})$.

Proposition 3.1.5

Let F be a field. Let $f \in F[x]$. Then any two splitting fields of f are isomorphic.

3.2 Normal Extensions

Definition 3.2.1 (Normal Extensions)

Let F be a field. Let K/F be a field extension. We say that K is a normal extension of F if there exists a set $S \subseteq F[x]$ such that K is the splitting field of all $f \in S$.

Proposition 3.2.2

Let F be a field. Let $S \subseteq F[x]$ be a subset. Then there exists a field extension K/F such that K is the splitting field of all $f \in S$.

Proposition 3.2.3

Let F be a field. Let K/F be an algebraic field extension. Then the following are equivalent.

- K/F is a normal extension.
- If $f \in F[x]$ is irreducible and K contains a root of f , then f splits over K .
- If $\varphi : K \rightarrow \bar{F}$ is an F -homomorphism, then $\varphi(K) = K$.

Proposition 3.2.4

Let F be a field. Let K/F be a field extension. Then the following are true.

- If L/F and K/L are field extensions and K/F is a normal extension, then K/L is a normal extension.
- If $F < L, M < K$ are field extensions such that $F < L, M$ are normal extensions, then LM/F and $L \cap M/F$ are normal extensions.

Proposition 3.2.5

Let F be a field. Let K/F be a finite field extension. Then K/F is normal if and only if it is the splitting field of some $f \in F[x]$.

Definition 3.2.6 (Normal Closure)

Let F be a field. Let K/F be an algebraic extension. Let L/K be an extension. We say that L is the normal closure of K/F if the following are true.

- L/K is a normal extension.
- If $K < M < L$ is a field extension such that M/F is normal, then $L = M$.

3.3 Separable Extensions**Definition 3.3.1 (Separable Polynomials)**

Let F be a field.

- Let $f \in F[x]$ be irreducible. We say that f is separable over F if f has no repeated roots in any splitting field of F .
- Let $f \in F[x]$ be a polynomial. We say that f is separable over F if all its irreducible factors are separable.

Proposition 3.3.2

Let F be a field. Let $f \in F[x]$ be non-constant irreducible polynomial. Then the following are true.

- If $\text{char}(F) = 0$, then f is separable over F .
- If $\text{char}(F) = p \neq 0$, then f is separable over F if and only if $f \notin F[x^p]$.

Definition 3.3.3 (Separable Extensions)

Let F be a field. Let K/F be a field extension. We say that K/F is a separable extension if $\min(F, a)$ is separable over F for all $a \in K$.

Example 3.3.4

Let F be a field. Let K/F be an algebraic field extension. If $\text{char}(F) = 0$, then K/F is a separable extension.

Proposition 3.3.5

Let F be a field. Let K/F and L/K be field extensions. Then L/F is separable if and only if L/K and K/F are separable.

Proposition 3.3.6

Let F be a field. Let K/F be a finitely generated field extension given by $K = F(a_1, \dots, a_n)$ for some $a_1, \dots, a_n \in K$. If each $F(a_i)/F$ is separable, then K/F is separable.

Theorem 3.3.7 (Primitive Element Theorem)

Let F be a field. Let K/F be a field extension. If K/F is finite and separable, then it is simple.

Definition 3.3.8 (Separable Closure)

Let F be a field. Let K/F be a field extension. Define the separable closure of F to be

$$S_{K/F} = \{\alpha \in K \mid \min(F, \alpha) \text{ is separable over } F\}$$

Definition 3.3.9 (Separable and Inseparable Degrees)

Let K/F be a finite field extension. Define the separable degree of K/F to be

$$[K : F]_{\text{sep}} = [S_{K/F} : F]$$

Define the inseparable degree of K/F to be

$$[K : F]_{\text{insep}} = [K : S_{K/F}]$$

Lemma 3.3.10

Let K/F be a finite field extension. Then we have

$$[S_{K/F} : F] = \text{Emb}_F(K, \overline{K})$$

Lemma 3.3.11

Let $L/K/F$ be finite field extensions. Then the following are true.

- $[L : F]_{\text{insep}} = [L : K]_{\text{insep}} [K : F]_{\text{insep}}$
- $[L : F]_{\text{sep}} = [L : K]_{\text{sep}} [K : F]_{\text{sep}}$

3.4 Perfect Fields

Definition 3.4.1 (Perfect Fields)

Let F be a field. We say that F is perfect if all its finite extensions are separable.

Lemma 3.4.2

Let F be a field. Then we have $\overline{F} = S_{\overline{F}/F}$.

Example 3.4.3

Let F be a field.

- If $\text{char}(F) = 0$, then F is perfect.
- If $\overline{F} = F$, then F is perfect.

4 The Galois Correspondence

4.1 Galois Extensions

Definition 4.1.1 (Galois Extension)

Let F be a field. Let K/F be an algebraic field extension. We say that K/F is a Galois extension if K is the splitting field of a separable polynomial with coefficients in F .

Note: We sometimes write $\text{Gal}(L/K) = \text{Aut}_K(L)$ when L/K is a Galois extension.

Proposition 4.1.2 (Equivalent characterization of Galois Extension)

Let F be a field. Let K/F be an algebraic field extension. Then K/F is a Galois extension if and only if $\text{Fix}_F(\text{Aut}_F(K)) = F$.

Proposition 4.1.3 (Equivalent characterization of Finite Galois Extension)

Let K/F be a finite field extension. Then the following are equivalent.

- K/F is a Galois extension.
- K/F is a normal and separable extension.
- $[K : F] = |\text{Aut}_F(K)|$.
- $\text{Fix}_F(\text{Aut}_F(K)) = F$.

Proof

- (1) $\iff$ (3): Let $n = [K : F]$. We induct on n .

When $n = 1$, then $K = F$ and there is exactly one F -automorphism. So suppose that $n > 1$. Let K be the splitting field of $f \in F[x]$ with $\deg(f) = d > 1$.

Let $a \in K \setminus F$ a root of f . Let $m \in F[x]$ be the minimal polynomial of a . This is an irreducible factor of f by definition. Break up the field extension into $F < F(a) < K$. We handle $K/F(a)$ by induction and $F(a)/F$ by hand.

Step 1: $K/F(a)$ is Galois by lemma 4.1.3. So by induction, we have

$$[K : F(a)] = |\text{Gal}(K/F(a))|$$

Step 2: Since f is separable, it has distinct roots in K . Thus the minimal polynomial m also has distinct roots in L as it divides f . By property of the minimal polynomial, we have that $[F(a) : F] = \deg(m)$. By corollary 2.1.5, we have that $\deg(m) = |\text{Emb}_F(F(a), K)|$.

Step 3: For every automorphism $\phi : K \rightarrow K$, the restriction to $F(a)$ gives a map of sets

$$\text{res}_a : \text{Gal}(K/F) \rightarrow \text{Emb}_F(F(a), K)$$

defined by $\phi \mapsto \phi|_{F(a)}$. I claim that this map is surjective. Let $\iota : F(a) \rightarrow K \in \text{Emb}_F(F(a), K)$. Since K/F is normal, by corollary 3.3.5 there exists an F -automorphism σ of K that restricts to ι . Let $\tau \in \text{Gal}(L/K)$ be any other map that restricts to ι . The map $\tau^{-1} \circ \sigma : K \rightarrow K$ is the identity on $F(a)$, and thus $\tau^{-1} \circ \sigma \in \text{Gal}(K/F(a))$. Thus τ and σ lie in the same coset, and thus

$$\frac{|\text{Gal}(K/F)|}{|\text{Gal}(K/F(a))|} = \left| \frac{\text{Gal}(K/F)}{\text{Gal}(K/F(a))} \right| = |\text{Emb}_F(F(a), K)|$$

Step 4: Putting the equalities together, we have that

$$\begin{aligned} |\text{Gal}(K/F)| &= |\text{Gal}(K/F(a))| \cdot |\text{Emb}_F(F(a), K)| \\ &= [K : F(a)] \cdot [F(a) : F] \\ &= [K : F] \end{aligned}$$

Thus we are done. ■

Lemma 4.1.4 If L/F is a Galois extension and $F < K < L$ an intermediate field, then L/K is a Galois extension.

Proof Let L be the splitting field of a separable polynomial $f \in F[x]$. Then L is also a splitting field of f regarded as a polynomial in $K[x]$. The factorization of f clearly does not change since f splits over L which K is a subfield of. ■

Beware that while L/K is a Galois extension in the above lemma, K/F may not be a Galois extension.

4.2 The Inclusion Reversing Bijection

Theorem 4.2.1 (The Fundamental Theorem of Galois Theory)

Let F be a field. Let K/F be a finite Galois extension. Then there is an inclusion reversing bijection

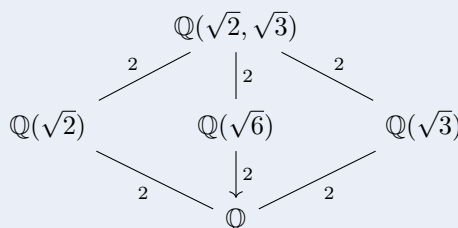
$$\left\{ \begin{array}{c} \text{Subgroups of} \\ \text{Aut}_F(K) \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Intermediary subfields} \\ F < L < K \end{array} \right\}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M , $M \mapsto \text{Aut}_L(K)$.

Example 4.2.2

The complete subfield lattice of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ is given by



Proof

Notice that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is the splitting field of $(x^2 - 2)(x^2 - 3)$ over $\mathbb{Q}$. It is also a finite extension and the polynomial is separable. Hence $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ is a Galois extension. Then we have

$$[\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3}))] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$$

Let $\sigma, \tau \in \text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3}))$ be defined by $\sigma(\sqrt{2}) = -\sqrt{2}$ and similarly $\tau(\sqrt{3}) = -\sqrt{3}$ while fixing the rest of the field. Clearly $|\sigma| = |\tau| = 2$ are order 2 elements of the group of automorphisms and they are different elements. Hence

$$\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2}, \sqrt{3})) = \{\text{id}, \sigma, \tau, \sigma \circ \tau\} \cong K_4$$

The only non-trivial proper subgroups of K_4 is given $\langle \sigma \rangle$, $\langle \tau \rangle$, $\langle \sigma \circ \tau \rangle$. By the fundamental theorem of Galois theory we have that the only three non-trivial proper intermediate fields of $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ are precisely the fixed fields of σ , τ and $\sigma \circ \tau$ respectively. Hence we are done. ■

Proposition 4.2.3 Let L/K be a Galois extension. Then the above bijection restricts to a bijection

$$\left\{ \begin{array}{c} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{c} \text{Normal Extensions} \\ K < M \end{array} \right\}$$

Proof Suppose that M/K is a normal extension. Then by proposition 2.3.7, we have that $\sigma(M) = M$ for any $M \in \text{Gal}(L/K)$. Suppose that $h \in \text{Gal}(L/M)$. Let $a \in M$ and $b = \sigma^{-1}(a)$. Notice that

$$\sigma h \sigma^{-1}(a) = \sigma(h(b)) = \sigma(b) = a$$

so that $\sigma h \sigma^{-1}$ fixes M . This means that $\sigma h \sigma^{-1} \in \text{Gal}(L/M)$.

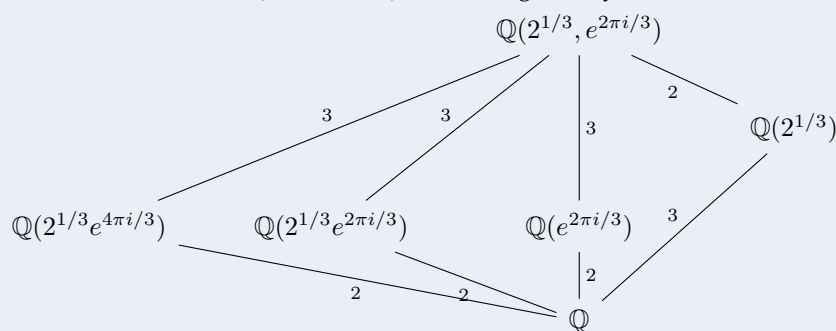
Conversely, suppose that $\text{Gal}(L/M) \trianglelefteq \text{Gal}(L/K)$. Let $a \in M$ has minimal polynomial $g \in K[x]$. We want to prove that g splits in M . Let b be a root of g . Then there exists some $\sigma \in \text{Gal}(L/K)$ such that $b = \sigma(a)$. For any $h \in \text{Gal}(L/M)$, we have that

$$\sigma h \sigma^{-1}(b) = \sigma h(a) = \sigma(a) = b$$

By normality, we have that $\sigma H \sigma^{-1} = H$ so that b is fixed by every element of H and so $b \in L^{\text{Gal}(L/M)} = M$, and so g splits over M , and M/K is normal. ■

Example 4.2.4

The complete subfield lattice of $\mathbb{Q}(2^{1/3}, e^{2\pi i/3})$ over $\mathbb{Q}$ is given by



Proof

Notice that $\mathbb{Q}(2^{1/3}, e^{2\pi i/3})$ is the splitting field of $x^3 - 2$ over $\mathbb{Q}$. It is also a finite extension and the polynomial is separable. Hence $\mathbb{Q}(2^{1/3}, e^{2\pi i/3})/\mathbb{Q}$ is a Galois extension. Then we have

$$[\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(2^{1/3}, e^{2\pi i/3}))] = [\mathbb{Q}(2^{1/3}, e^{2\pi i/3}) : \mathbb{Q}] = 6$$

Since any automorphism must permute the roots each minimal polynomial of $2^{1/3}$ and $e^{2\pi i/3}$, we deduce that any automorphism must send $2^{1/3}$ to $\{2^{1/3}, 2^{1/3}e^{2\pi i/3}, 2^{1/3}e^{4\pi i/3}\}$ and must send $e^{2\pi i/3}$ to $\{e^{2\pi i/3}, e^{4\pi i/3}\}$. This gives a total of six possibilities which matches the order of the group of automorphisms. Let σ be the automorphism defined by

$$\sigma = \begin{cases} 2^{1/3} \mapsto 2^{1/3}e^{2\pi i/3} \\ e^{2\pi i/3} \mapsto e^{2\pi i/3} \end{cases}$$

Let τ be the automorphism defined by

$$\tau = \begin{cases} 2^{1/3} \mapsto 2^{1/3} \\ e^{2\pi i/3} \mapsto e^{4\pi i/3} \end{cases}$$

Clearly $|\sigma| = 3$ and $|\tau| = 2$. It remains to compute the group structure. Notice that $\sigma\tau = \tau\sigma^2$ and so

$$\text{Aut}_{\mathbb{Q}}(\mathbb{Q}(2^{1/3}, e^{2\pi i/3})) = \{\text{id}, \sigma, \sigma^2, \tau, \sigma\tau, \sigma^2\tau\} \cong S_3$$

The non-trivial proper subgroups of S_3 are precisely $\langle \sigma \rangle, \langle \tau \rangle, \langle \sigma\tau \rangle, \langle \sigma^2\tau \rangle$. By the fundamental theorem of Galois theory we have that the only three non-trivial proper intermediate fields of $\mathbb{Q}(2^{1/3}, e^{2\pi i/3})/\mathbb{Q}$ are precisely the fixed fields of $\sigma, \tau, \sigma\tau, \sigma^2\tau$ respectively. Hence we are done. ■

Proposition 4.2.5 Let $K < M < L$ be field extensions such that L/K is Galois and M/K is normal. Then we have

$$\text{Aut}_K(M) \cong \frac{\text{Aut}_K(L)}{\text{Fix}_M(\text{Aut}_K(L))}$$

Moreover, the following are true.

- $[L : M] = |\text{Fix}_M(\text{Aut}_K(L))|$
- $[M : K] = \left| \frac{\text{Aut}_K(L)}{\text{Fix}_M(\text{Aut}_K(L))} \right|$

Let L/K be a Galois extension. We summarize what we know about $\text{Gal}(L/K) = \text{Aut}_K(L)$.

- $|\text{Aut}_K(L)| \leq [L : K]$.
- $[L : \text{Fix}_K(H)] = |H|$ where $H \leq \text{Aut}_K(L)$.

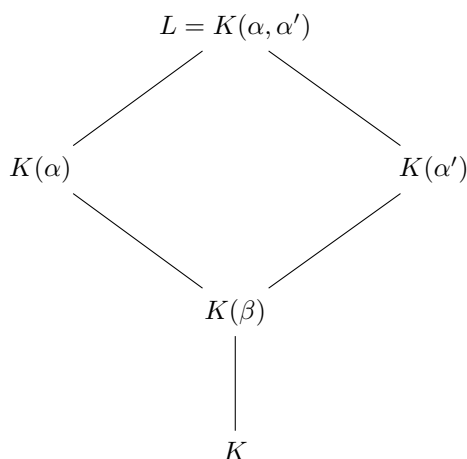
4.3 Application: Biquadratic Extensions

As an immediate application of the fundamental theorem of Galois theory, we apply the inclusion reversing bijection to biquadratic extensions. The set up is as follows:

Let K be a field such that $\text{char}(K) \neq 2$. Let $a, b \in K$ such that $b(a^2 - b) \neq 0$ and b is not a square in K . Consider

$$f(x) = x^4 - 2ax^2 + (a^2 - b) \in K[x]$$

Write $\beta^2 = b$. Then it is easy to see that $x^2 - a = \pm\beta$ so that the roots of f are $\alpha, -\alpha, \alpha', -\alpha' \in L$ the splitting field of K by taking quadratic extensions (degree 2 extensions) where $\alpha^2 = a + \beta$ and $(\alpha')^2 = a - \beta$. In particular, $L = K(\alpha, \alpha')$. It is easy to see that the four roots are four distinct elements. Together with the fact that $\deg(f) = 4$ shows that L/K is a Galois extension (It may be easier to show that $Df = 4x(x^2 - a)$ is coprime with f). We can see L/K in a tower:



Notice that β is omitted in the bigger fields since $\beta = \alpha^2 - a = a - (\alpha')^2$. Each inclusion in the tower is either an equality or a degree 2 extension. Thus $[L : K] = 2, 4$ or 8 . We need a lemma.

Lemma 4.3.1 Let K be a field. Let $u, v \in M$. If u is a square in $M(\sqrt{v})$ then either u or uv is a square in M .

Proof This follows since

$$(c^2 + d^2v) + 2cd\sqrt{v} = (c + d\sqrt{v})^2 = u \in M$$

would imply that $cd = 0$. If $d = 0$, then $u = c^2$. If $c = 0$, then $u = d^2v$ so that $uv = (dv)^2$ and we are done. ■

Proposition 4.3.2 Let K be a field such that $\text{char}(K) \neq 2$. Let $a, b \in K$ such that $b(a^2 - b) \neq 0$ and b is not a square in K . Consider

$$f(x) = x^4 - 2ax^2 + (a^2 - b) \in K[x]$$

Write $\beta^2 = b$. Let L/K be the splitting field of f . Denote the four distinct roots of f by $\alpha, -\alpha, \alpha', -\alpha' \in L$ for $\alpha^2 = a + \beta$ and $(\alpha')^2 = a - \beta$.

- If $a^2 - b$ is not a square in K , then $[K(\alpha) : K] = [K(\alpha') : K] = 4$

- If neither $a^2 - b$ nor $b(a^2 - b)$ is a square in K , then $[L : K] = 8$.

Proof Suppose that $a^2 - b$ is not a square in K . Since $b \in K$ is not a square, we have $[K(\beta) : K] = 2$. It remains to show that $[K(\alpha) : K(\beta)] = 2$. Suppose for a contradiction that $\alpha = c + d\beta \in K(\beta)$ for some $c, d \in K$. Then we have that

$$a + \beta = \alpha^2 = (c + d\beta)^2 = (c^2 + d^2b) + 2cd\beta$$

and so $a - \beta = (c - d\beta)^2$ is also a square in $K(\beta)$. Thus

$$a^2 - b = (a + \beta)(a - \beta) = ((c + d\beta)(c - d\beta))^2$$

is a square in K , a contradiction. Thus we have that $\alpha \notin K(\beta)$ and $[K(\alpha) : K] = 4$. Similarly, we have that $\alpha' \notin K(\beta)$ and thus $[K(\alpha') : K] = 4$.

Now suppose further that $b(a^2 - b)$ is not a square in K . We need to show that $\alpha' \notin K(\alpha)$. Suppose for a contradiction that $\alpha' \in K(\alpha)$. Writing $\alpha' = c + d\alpha$ with $c, d \in K(\beta)$, we have that

$$a - \beta = (\alpha')^2 = (c + d\alpha)^2 = (c^2 + d^2(a + \beta)) + 2cd\alpha$$

Thus $cd = 0$ since $\{1, \alpha\}$ is a basis for $K(\alpha)/K(\beta)$. Since $\alpha' \in K(\beta)$ by the first part of the proof, we have $d \neq 0$. Thus $c = 0$ and $a - \beta = d^2(a + \beta)$. Therefore

$$a^2 - b = (a - \beta)(a + \beta) = (d(a + \beta))^2$$

is a square in $K(\beta)$. By the above lemma, either $a^2 - b$ or $b(a^2 - b)$ is a square in K , both of which are contradictions and so we conclude. ■

Lemma 4.3.3 Let K be a field such that $\text{char}(K) \neq 2$. Let $a, b \in K$ such that $b(a^2 - b) \neq 0$ and b is not a square in K . Consider

$$f(x) = x^4 - 2ax^2 + (a^2 - b) \in K[x]$$

Write $\beta^2 = b$. Let L/K be the splitting field of f . Denote the four distinct roots of f by $\alpha, -\alpha, \alpha', -\alpha' \in L$ for $\alpha^2 = a + \beta$ and $(\alpha')^2 = a - \beta$.

Suppose further that $[K(\alpha) : K] = [K(\alpha') : K] = 4$.

- $q(x) = x^2 - (a + \beta)$ is the minimal polynomial of α over $K(\beta)$, and $q'(x) = x^2 - (a - \beta)$ is that of α'
- Any $\sigma \in \text{Gal}(L/K)$ can only map β, α, α' in one of the following 8 ways:

	1	2	3	4	5	6	7	8
$\sigma(\beta)$	β	β	β	β	$-\beta$	$-\beta$	$-\beta$	$-\beta$
$\sigma(\alpha)$	α	α	$-\alpha$	$-\alpha$	α'	α'	$-\alpha'$	$-\alpha'$
$\sigma(\alpha')$	α'	$-\alpha'$	α'	$-\alpha'$	α	$-\alpha$	α	$-\alpha$

(No claim in existence of these maps)

Proof

- The assumption on degree implies that each step in the tower has degree 2, so that it ensures the irreducibility of both.
- By corollary 1.1.5, either $\sigma(\beta) = \beta$ or $-\beta$. If $\sigma(\beta) = \beta$, then σ fixes $K(\beta)$ so that by the same corollary, σ must permute the roots of q and the roots of q' respectively. If instead $\sigma(\beta) = -\beta$, then notice that $\sigma(q) = q'$ and so σ must exchange the roots of q and q' in some order.

And so we are done. ■

Theorem 4.3.4 Let K be a field such that $\text{char}(K) \neq 2$. Let $a, b \in K$ such that $b(a^2 - b) \neq 0$ and b is not a square in K . Consider

$$f(x) = x^4 - 2ax^2 + (a^2 - b) \in K[x]$$

Let L/K be the splitting field of f . Then the following completely classifies $\text{Aut}_K(L)$.

- If $[L : K] = 8$, then $\text{Aut}_K(L) \cong D_8$
- If $\sqrt{b(a^2 - b)} \in K$, then $[L : K] = 4$ and $\text{Aut}_K(L) \cong C_4$
- If $\sqrt{a^2 - b} \in K$ and $[L : K] = 4$ then $\text{Aut}_K(L) \cong C_2 \times C_2$
- Otherwise, $[L : K] = 2$ and $\text{Aut}_K(L) \cong C_2$

Proof Write $\beta^2 = b$. Denote the four distinct roots of f by $\alpha, -\alpha, \alpha', -\alpha' \in L$ for $\alpha^2 = a + \beta$ and $(\alpha')^2 = a - \beta$.

•

- Notice that $(\beta\alpha\alpha')^2 = b(a^2 - b)$ so that $\beta\alpha\alpha' \in K$ by the case assumption. Thus $\alpha' \in K(\alpha, \beta) = K(\alpha)$, where $L = K(\alpha) = K'(\alpha)$. Since b is not a square implies that $a^2 - b$ is not a square, we have $[L : K] = 4$ by proposition 3.4.2.

Now f is irreducible in K . Indeed, it has no roots in K and so we just have to check that f is not a product of two irreducible quadratics. By a computation, we can show that all pairwise products of linear factors $x - \alpha, x - \alpha', x + \alpha$ and $x + \alpha'$ are not factors of f in K . Since f is irreducible, there exists some $\sigma \in \text{Gal}(L/K)$ with $\sigma(\alpha) = \alpha'$. Lemma 3.4.3 implies that $\sigma(\beta) = -\beta$ and since $\beta\alpha\alpha' \in K$ is fixed by σ , it follows that $\sigma(\alpha') = -\alpha$. Thus σ has order 4.

But we also know that $\text{Gal}(L/K) = 4$ since L/K is Galois. So $\text{Gal}(L/K) = \langle \sigma \rangle$ is a cyclic group of order 4.

- If $\sqrt{a^2 - b} \in K$, then $a^2 - b = (\alpha\alpha')^2 \in K$ so that $\alpha\alpha' \in K$. Then we have $\alpha' \in K(\alpha)$ so that $L = K(\alpha) = K(\alpha')$. By the Galois correspondence, we have that $\text{Gal}(L/K) = 4$ since L/K is Galois. There is an element $\nu \in \text{Gal}(K(\beta)/K)$ such that $\nu(\alpha) = -\alpha$ and $\nu(\beta) = \beta$ by lemma 3.4.3. Since $\alpha\alpha'$ is fixed by ν , we also have $\nu(\alpha') = -\alpha'$. There is also an element of $\text{Gal}(K(\beta)/K)$ that maps β to $-\beta$. This must lift to an element $\rho \in \text{Gal}(K(\beta)/K)$. This automorphism takes α to one of the roots of $\rho(q(x)) = q'(x)$ as in lemma 3.4.3. If $\rho(\alpha) = \alpha'$, then we have 4 distinct automorphisms

$$\text{id}, \nu, \rho, \nu\rho \in \text{Gal}(L/K)$$

where $\nu\rho(\alpha) = -\alpha'$ and so is distinct from the first three. If instead $\rho(\alpha) = -\alpha'$ then we will still get the four automorphisms. So fix notation with $\rho(\alpha) = \alpha'$ so that

$$\text{Gal}(L/K) = \langle \nu \rangle \times \langle \rho \rangle \cong C_2 \times C_2$$

■

5 Norm and Trace

5.1 The Norm and Trace Map

Definition 5.1.1 (The Multiplication Map) Let K be a field. Let $\alpha \in K$. Define the multiplication map by α to be the map $M_\alpha : K \rightarrow K$ where $\beta \mapsto \alpha\beta$ for all $\beta \in K$.

Proposition 5.1.2 Let K/F be a field extension. Then the characteristic polynomial χ_{M_α} of the multiplication map $M_\alpha : K \rightarrow K$ is equal to the minimal polynomial $\min_{K,\alpha}$.

Proof Let $\chi_{M_\alpha} = \sum_{k=0}^n a_n x^n$ be the characteristic polynomial of M_α . In particular we must have $a_n = 1$ because characteristic polynomials are monic. By the Cayley-Hamilton theorem, $\chi_{M_\alpha}(M_\alpha) = 0$, where both sides of the equation is considered as functions $K \rightarrow K$. Now evaluate the function at 1_K to obtain $\sum_{k=0}^n a_n \alpha^n = 0$. Thus $\min_{K,\alpha}$ divides χ_{M_α} . Now $\min_{K,\alpha}$ is monic and has degree $\deg(\min_{K,\alpha}) = [K : F]$. But also $\deg(\chi_{M_\alpha}) = [K : F]$. Thus we conclude that $\min_{K,\alpha} = \chi_{M_\alpha}$. ■

Proposition 5.1.3 Let L/K be an extension of number fields. Let $\alpha \in K$. Then

$$\chi_{L,M_\alpha} = \chi_{K,M_\alpha}^{[L:K]}$$

Definition 5.1.4 (The Trace Map) Let L/K be a finite field extension. Define the trace map $\text{Tr}_{L/K} : L \rightarrow K$ to be the map

$$\text{Tr}_{L/K}(\alpha) = \text{tr}(M_\alpha)$$

where $M_\alpha : L \rightarrow L$ is the multiplication map defined by $\beta \mapsto \alpha\beta$.

Lemma 5.1.5

Let L/K be a finite field extension. Then the following are true.

- $\text{Tr}_{L/K}$ is K -linear.
- If $a \in K$, then $\text{Tr}_{L/K}(a) = [L : K] \cdot a$.

Definition 5.1.6 (The Norm Map) Let L/K be a finite field extension. Define the trace map $N_{L/K} : L \rightarrow K$ to be the map

$$N_{L/K}(\alpha) = \det(M_\alpha)$$

where $M_\alpha : L \rightarrow L$ is the multiplication map defined by $\beta \mapsto \alpha\beta$.

Lemma 5.1.7

Let L/K be a finite field extension. Then the following are true.

- $N_{L/K} : L^\times \rightarrow K^\times$ is a group homomorphism.
- If $a \in K$, then $\text{Tr}_{L/K}(a) = a^{[L:K]}$.

Corollary 5.1.8 Let $L/K/F$ be an extension of number fields. Then the following are true.

- $\text{Tr}_{L/F} = [L : K] \text{Tr}_{K/F}$
- $N_{L/F} = N_{K/F}^{[L:K]}$

Theorem 5.1.9

Let K/F be a finite field extension. Then the following are true.

- The norm of K over F is given by

$$N_{K/F} = \left(\prod_{\sigma \in \text{Emb}_F(K, \overline{K})} \sigma \right)^{[K:F]_{\text{insep}}}$$

- The trace of K over F is given by

$$\mathrm{Tr}_{K/F} = [K : F]_{\mathrm{insep}} \left(\sum_{\sigma \in \mathrm{Emb}_F(K, \overline{K})} \sigma \right)$$

5.2 The Norm and Trace of Galois Extensions

Corollary 5.2.1

Let K/F be a Galois extension. Then the following are true.

- The norm of K over F is given by

$$N_{K/F} = \prod_{\sigma \in \mathrm{Gal}(K/F)} \sigma$$

- The trace of K over F is given by

$$\mathrm{Tr}_{K/F} = \sum_{\sigma \in \mathrm{Gal}(K/F)} \sigma$$

Corollary 5.2.2

Let $L/K/F$ be a finite field extensions. Then the following are true.

- $\mathrm{Tr}_{L/F} = \mathrm{Tr}_{K/F} \circ \mathrm{Tr}_{L/K}$.
- $N_{L/F} = N_{K/F} \circ N_{L/K}$.

6 Finite Field Theory

6.1 Existence and Uniqueness of Finite Fields

Proposition 6.1.1 If K is a finite field, then it has $\text{char}(K) = p$ a prime, and $|K| = p^n$ for some $n \in \mathbb{N} \setminus \{0\}$.

Proof Since K is finite, there is no field homomorphism $\mathbb{Z} \rightarrow K$ since such a homomorphism must be injective. This means that the prime subfield of K must be $\mathbb{F}_p$ for some prime p . So K is a finite dimensional vector space over $\mathbb{F}_p$, and after choosing any basis, its elements are simply vectors $(a_1, \dots, a_n)$ where $a_i \in \mathbb{F}_p$, where $n = [K : \mathbb{F}_p]$, and there are obviously p^n of these vectors. ■

Lemma 6.1.2 Let $\mathbb{F}$ be any field. A finite subgroup of $\mathbb{F}^\times$ is cyclic.

Proposition 6.1.3 Every finite field is isomorphic to $\frac{\mathbb{F}_p[x]}{(f)}$ where p is a prime number and $f \in \mathbb{F}_p[x]$ is an irreducible polynomial.

6.2 Galois Theory for Finite Fields

Theorem 6.2.1 Let p be a prime number and $n \geq 1$. Let $q = p^n$. Then the splitting field L of

$$f(x) = x^q - x \in \mathbb{F}_p[x]$$

is a field with p^n elements. Moreover, $L/\mathbb{F}_p$ is a Galois extension.

Proof Let $L/\mathbb{F}_p$ be the splitting field for f . Since the formal derivative of f is $Df = -1$, f is separable over L with n distinct roots. Thus $L/\mathbb{F}_p$ is a Galois extension. Let $M \subset L$ be the set of roots of f in L , then $M = \{\alpha \in L \mid \alpha^q = \alpha\}$ and $|M| = q$. It is easy to check that M is a field in its own right so that $M = L$ by minimality of the splitting field. ■

Proposition 6.2.2 Any two fields of order p^n are isomorphic.

Proof By the above theorem, one such field of order n is the splitting field L of the polynomial $f(x) = x^q - x \in \mathbb{F}_p[x]$. Suppose $N/\mathbb{F}_p$ is another field of $q = p^n$ elements. Consider the set N^* of non-zero elements as a finite group under multiplication. We must have $\beta^{q-1} = 1$ for any $\beta \in N^*$ by Lagrange's theorem. Multiplying by β , we have that $\beta^q = \beta$ for all $\beta \in N$. That is, f splits in N . Since $|N|$ is the number of roots of f in any splitting field, N is the splitting field for f over $\mathbb{F}_p$. Since splitting fields are unique up to isomorphism, we conclude. ■

We denote this unique field up to isomorphism $\mathbb{F}_{p^n}$

Definition 6.2.3 (Frobenius Map) Let K be a field of characteristic $p > 0$. The Frobenius map is the homomorphism

$$\varphi_p : K \rightarrow K$$

defined by $\alpha \mapsto \alpha^p$.

Lemma 6.2.4 The Frobenius map is a field homomorphism.

Proof We have that

$$\begin{aligned}
 \varphi(a+b) &= (a+b)^p \\
 &= \sum_{k=0}^p \binom{p}{k} a^k b^{p-k} \\
 &= a^p + b^p \\
 &= \varphi(a) + \varphi(b)
 \end{aligned}$$

and

$$\varphi(ab) = (ab)^p = a^p b^p = \varphi(a)\varphi(b)$$

and so we conclude. ■

Proposition 6.2.5 Let K be a field of characteristic $p > 0$. Let $\varphi_p : K \rightarrow K$ be Frobenius map. Then the following are true.

- $F_p = \{a \in K \mid \varphi_p(a) = a\}$
- If K is a finite field, then φ_p is surjective and $\varphi_p \in \text{Aut}_{F_p}(K)$. In this case, it has fixed field $K^{\varphi_p} = F_p$

Proof The set of elements on which φ is the identity is certainly a subfield of K since φ is a field homomorphism which contains the prime subfield. At the same time, elements such that $\varphi_p(a) = a$ can be interpreted as the roots in K of the polynomial $x^p - x \in F_p[x]$. This is not the zero polynomial and has at most p roots. F_p has p roots and so they must be all of them.

If K is a finite field, then it is a finite dimensional vector space over F_p , and φ_p is an injective linear map, thus φ_p is an isomorphism. In particular, the first part of the proposition shows that φ_p fixes F_p , so it is an F_p -automorphism. Its fixed field is given precisely by the set $\{a \in K \mid \varphi_p(a) = a\}$. By the first part of the proof, this is exactly F_p . ■

Theorem 6.2.6 Let p be prime. Denote $\varphi_p : F_{p^n} \rightarrow F_{p^n}$ the Frobenius map. Then

$$\text{Gal}(F_{p^n}/F_p) = \langle \varphi_p \rangle \cong \mathbb{Z}/n\mathbb{Z}$$

is a cyclic group of order n generated by the Frobenius map.

Proof The extension F_{p^n}/F_p is Galois by theorem 4.1.2, so

$$\text{Gal}(F_{p^n}/F_p) = [F_{p^n} : F_p] = n$$

by theorem 3.2.2. When $n = 1$, the case is trivial. Suppose that $n > 1$. $\varphi_p \in \text{Gal}(F_{p^n}/F_p)$ is a non trivial element by the above proposition. φ_p has order at most n . We show that the order is exactly n . If $\varphi_p^m = \text{id}$ for some $m \leq n$, then for every $\alpha \in K$,

$$\alpha^{p^m} = \varphi_p^m(\alpha) = \alpha$$

and so $g(x) = x^{p^m} - x$ has p^m solutions in F_{p^n} . Then $p^m = \deg(g) \geq p^n$, so that $m = n$. ■

Notice that it follows that F_{p^m} is a subfield of F_{p^n} if and only if m divides n .

7 Cyclotomic Fields and Polynomials

7.1 Cyclotomic Polynomials

Lemma 7.1.1 Let $d, n \in \mathbb{N} \setminus \{0\}$ where $d|n$. Then $x^d - 1$ divides $x^n - 1$ as elements of the polynomial ring $\mathbb{Z}[x]$.

Lemma 7.1.2 Let $n > 0$ be an integer and for $0 \leq i < n$, define $d_i = \gcd(n, i)$. Then the following are true regarding the n -th roots of unity $\{\zeta_n^0, \zeta_n^1, \dots, \zeta_n^{n-1}\}$.

- ζ_n^i has order n/d_i
- ζ_n^i is a primitive (n/d_i) th root of unity
- ζ_n^i is a primitive n th root of unity if and only if $d_i = 1$

Definition 7.1.3 (Cyclotomic Polynomials) Let $n \in \mathbb{N} \setminus \{0\}$. Define the n th cyclotomic polynomial as

$$\Phi_n(x) = \prod_{\substack{0 \leq i < n \\ \gcd(n, i) = 1}} (x - \zeta_n^i)$$

In other words the polynomial is the product over all linear factors of primitive n th roots of unity.

Proposition 7.1.4 The cyclotomic polynomials factorizes in $\mathbb{C}[x]$ into

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

Proposition 7.1.5 For each $n \in \mathbb{N}$, $\Phi_n(x)$ is monic and has coefficients in $\mathbb{Z}$.

7.2 Cyclotomic Fields

Definition 7.2.1 (n-th Cyclotomic Field) Let $n \in \mathbb{N}$ and let $\zeta_n = e^{2\pi i/n}$ denote the n th root of unity. Define the n -th cyclotomic field to be $\mathbb{Q}(\zeta_n)$.

Proposition 7.2.2 Let p be a prime number. Then the minimal polynomial of ζ_p is the n th cyclotomic polynomial Φ_n . In particular, we have that

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

This is exactly the Euler totient function $\phi(n)$ in number theory.

8 Radial and Soluble Extensions

8.1 Radical Extensions

Roughly speaking, a polynomial is soluble by radicals if its roots can be expressed using n th roots. This motivates the definition of radical extensions and soluble extensions.

Definition 8.1.1 (Radical Extensions) A field extension L/K is radical if there is a sequence of subfields

$$K = F_0 < F_1 < \cdots < F_n = L$$

such that $F_k = F_{k-1}(\alpha_i)$ where $\alpha_i^{n_i} \in F_{k-1}$ for some $n_i > 0$.

Proposition 8.1.2 Suppose that $\text{char}(K) = 0$. If L/K is a radical extension, then there exists a finite extension M/L so that M/K is both radical and Galois.

Definition 8.1.3 (Soluble Extensions) A field extension L/K is soluble if L is contained in a field M with M/K radical. A polynomial $f \in K[x]$ is soluble by radicals if its splitting field L is soluble.

Thus, if f is soluble by radicals, every element of its splitting field has an expression that involves only n th roots, field operations and elements of K .

Proposition 8.1.4 Let K be a field such that $\text{char}(K) = 0$. If L/K is a radical extension, then there is a finite extension M/L so that M/K is both radical and Galois.

9 Transcendental Extensions

9.1 Transcendence Basis

Definition 9.1.1 (Algebraically Independent) Let K/F be a field extension. We say that a set $S \subset K$ is algebraically independent if for any finite set $\{a_1, \dots, a_n\} \subseteq S$, there exists no $f \in F[x_1, \dots, x_n]$ such that $f(a_1, \dots, a_n) = 0$. We say that S is algebraically dependent otherwise.

An algebraically independent set of elements behave similarly to variables in a polynomial ring. The following statement demonstrates this.

Lemma 9.1.2 Let K/F be a field extension and let $\{a_1, \dots, a_n\} \subset K$ be an algebraically independent set of elements. Then $F[a_1, \dots, a_n]$ is F -isomorphic to $F[x_1, \dots, x_n]$. Moreover $F(a_1, \dots, a_n)$ is F -isomorphic to $F(x_1, \dots, x_n)$.

Proposition 9.1.3 Let K/F be a field extension. Let $t_1, \dots, t_n \in K$. Then the following statements are equivalent.

- $\{t_1, \dots, t_n\}$ is algebraically independent over F
- For each i , t_i is transcendental over $F(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_n)$
- For each i , t_i is transcendental over $F(t_1, \dots, t_n)$.

Definition 9.1.4 (Transcendence Basis) Let K/F be a field extension. A subset S of K is a transcendence basis for K/F if S is algebraically independent over F and if K is algebraic over $F(S)$.

Lemma 9.1.5 Let K be a field extension of F . Let $S \subseteq K$ be algebraically independent over F . If $t \in K$ is transcendental over $F(S)$, then $S \cup \{t\}$ is algebraically independent over F .

We can now prove that every field extension has a transcendence basis.

Proposition 9.1.6 Every field extension has a transcendence basis.

Similar to bases in vector spaces, the cardinality of any two transcendence basis is the same.

Theorem 9.1.7 Let K be a field extension of F . If S and T are transcendence bases for K/F , then $|S| = |T|$.

Again, recall that linearly independent sets can be extended to a basis and spanning sets can be reduced to a basis in vector spaces. There is similar analogue for transcendence basis.

Theorem 9.1.8 Let K/F be a field extension. Then the following are true.

- If $T \subseteq K$ is such that $K/F(T)$ is algebraic, then T contains a transcendence basis.
- If $S \subseteq K$ is algebraically independent, then S is contained in a transcendence basis of K/F .

9.2 Transcendence Degree

The following definition is well defined since the cardinality of transcendence basis is an invariant of field extensions.

Definition 9.2.1 (Transcendence Degree) Let K/F be a field extension. Define the transcendence degree $\text{trdeg}(K/F)$ of K/F to be the cardinality of any transcendence basis of K/F .

Proposition 9.2.2 Let K/F and L/K be field extensions. Then

$$\text{trdeg}(L/F) = \text{trdeg}(L/K) + \text{trdeg}(K/F)$$

9.3 Linear Disjointness